

Лабораторная работа № 6

Настройка пропускной способности глобальной сети с помощью Token Bucket Filter

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Основной целью работы является знакомство с принципами работы дисциплины очереди Token Bucket Filter, которая формирует входящий/исходящий трафик для ограничения пропускной способности, а также получение навыков моделирования и исследования поведения трафика посредством проведения интерактивного и воспроизводимого экспериментов в Mininet.

Выполнение лабораторной работы

```
root@mininet-vm:/home/mininet# ifconfig
h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> ntu 1500
        inet 10.0.0.1 netmask 255.0.0.0 broadcast 10.255.255.255
        ether ba:ec:0b:1a:34:30 txqueuelen 1000 (Ethernet)
        RX packets 0 bytes 0 (0.0 B)
        RX errors 0 dropped 0 overruns 0 frame 0
        TX packets 0 bytes 0 (0.0 B)
        TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> ntu 65536
        inet 127.0.0.1 netmask 255.0.0.0
        loop txqueuelen 1000 (Local Loopback)
        RX packets 1502 bytes 108308 (108.3 KB)
        RX errors 0 dropped 0 overruns 0 frame 0
        TX packets 1502 bytes 108308 (108.3 KB)
        TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet# ping -c 4
ping: usage error: Destination address required

root@mininet-vm:/home/mininet# ifconfig
h2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> ntu 1500
        inet 10.0.0.2 netmask 255.0.0.0 broadcast 10.255.255.255
        ether 0e:c1:4a:61:e0:a4 txqueuelen 1000 (Ethernet)
        RX packets 0 bytes 0 (0.0 B)
        RX errors 0 dropped 0 overruns 0 frame 0
        TX packets 0 bytes 0 (0.0 B)
        TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> ntu 65536
        inet 127.0.0.1 netmask 255.0.0.0
        loop txqueuelen 1000 (Local Loopback)
        RX packets 1408 bytes 174320 (174.3 KB)
        RX errors 0 dropped 0 overruns 0 frame 0
        TX packets 1408 bytes 174320 (174.3 KB)
        TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet#

switch: s1 (root@mininet-vm)

root@mininet-vm:/home/mininet# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> ntu 1500
        inet 192.168.56.8 netmask 255.255.255.0 broadcast 192.168.56.255
        ether 08:00:27:d1:db:d9 txqueuelen 1000 (Ethernet)
        RX packets 3181 bytes 783593 (783.5 KB)
        RX errors 0 dropped 0 overruns 0 frame 0
        TX packets 2148 bytes 343821 (343.8 KB)
        TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> ntu 1500
        inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
        ether 08:00:27:d1:8a:f1 txqueuelen 1000 (Ethernet)
        RX packets 418 bytes 40591 (40.5 KB)
        RX errors 0 dropped 0 overruns 0 frame 0
        TX packets 457 bytes 41372 (41.3 KB)
        TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> ntu 65536
        inet 127.0.0.1 netmask 255.0.0.0
        loop txqueuelen 1000 (Local Loopback)
        RX packets 1502 bytes 108308 (108.3 KB)
        RX errors 0 dropped 0 overruns 0 frame 0
        TX packets 1502 bytes 108308 (108.3 KB)
        TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

switch: s2 (root@mininet-vm)

root@mininet-vm:/home/mininet# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> ntu 1500
        inet 192.168.56.8 netmask 255.255.255.0 broadcast 192.168.56.255
        ether 08:00:27:d1:db:d9 txqueuelen 1000 (Ethernet)
        RX packets 3466 bytes 835695 (835.6 KB)
        RX errors 0 dropped 0 overruns 0 frame 0
        TX packets 2438 bytes 373303 (373.3 KB)
        TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> ntu 1500
        inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
        ether 08:00:27:d1:8a:f1 txqueuelen 1000 (Ethernet)
        RX packets 423 bytes 49041 (49.0 KB)
        RX errors 0 dropped 0 overruns 0 frame 0
        TX packets 462 bytes 41822 (41.8 KB)
        TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> ntu 65536
        inet 127.0.0.1 netmask 255.0.0.0
        loop txqueuelen 1000 (Local Loopback)
        RX packets 7938 bytes 949481 (949.4 KB)
        RX errors 0 dropped 0 overruns 0 frame 0
        TX packets 7938 bytes 949481 (949.4 KB)
        TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Выполнение лабораторной работы

```
root@mininet-vn:/home/nininet# ping -c 4 10.0.0.2
ping: usage error: Destination address required
root@mininet-vn:/home/nininet# ping -c 4 10.0.0.2
ping: command not found
root@mininet-vn:/home/nininet# ping -c 4 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=5.56 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.264 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.000 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.047 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 303ms
rtt min/avg/max/ndev = 0.047/1.487/5.559/2.352 ms
root@mininet-vn:/home/nininet# ping -c 4 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.035 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.067 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.061 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.058 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 304ms
rtt min/avg/max/ndev = 0.035/0.055/0.067/0.012 ms
root@mininet-vn:/home/nininet#

root@h2@mininet-vn#
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
loop txqueuelen 1000 (Local Loopback)
RX packets 1408 bytes 174320 (174.3 KB)
TX errors 0 dropped 0 overruns 0 frame 0
TX packets 1408 bytes 174320 (174.3 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vn:/home/nininet# ^C
root@mininet-vn:/home/nininet# ping -c 4 10.0.0.1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=2.54 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0.095 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=0.054 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0.055 ms

--- 10.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 303ms
rtt min/avg/max/ndev = 0.054/0.085/2.539/1.070 ms
```

Выполнение лабораторной работы

```
host: h1@mininet-vm
--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3048ms
rtt min/avg/max/ndev = 0.035/0.055/0.067/0.012 ms
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 60184 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer      Bitrate      Retrans      Cwnd
[ 7] 0.00-1.00 sec  1.32 GBytes  11.3 Gbits/sec  0          8.22 MBytes
[ 7] 1.00-2.01 sec  1.21 GBytes  10.4 Gbits/sec  1          5.84 MBytes
[ 7] 2.01-3.00 sec  1.14 GBytes  9.89 Gbits/sec  0          6.07 MBytes
[ 7] 3.00-4.01 sec  1.40 GBytes  11.9 Gbits/sec  1          6.28 MBytes
[ 7] 4.01-5.00 sec  1.45 GBytes  12.6 Gbits/sec  0          6.48 MBytes
[ 7] 5.00-6.00 sec  1.39 GBytes  11.9 Gbits/sec  0          6.62 MBytes
[ 7] 6.00-7.00 sec  1.65 GBytes  14.1 Gbits/sec  0          6.80 MBytes
[ 7] 7.00-8.00 sec  1.81 GBytes  15.6 Gbits/sec  0          6.95 MBytes
[ 7] 8.00-9.00 sec  1.74 GBytes  14.9 Gbits/sec  0          7.08 MBytes
[ 7] 9.00-10.01 sec 1.25 GBytes  10.7 Gbits/sec  0          7.22 MBytes
[ ID] Interval      Transfer      Bitrate      Retrans
[ 7] 0.00-10.01 sec 14.3 GBytes  12.3 Gbits/sec  2
[ 7] 0.00-10.01 sec 14.3 GBytes  12.3 Gbits/sec
iperf Done.
root@mininet-vm:/home/mininet#

host: h2@mininet-vm
Server listening on 5201
Accepted connection from 10.0.0.1, port 60182
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 60184
[ ID] Interval      Transfer      Bitrate      Retrans      Cwnd
[ 7] 0.00-1.00 sec  1.32 GBytes  11.3 Gbits/sec  0          8.22 MBytes
[ 7] 1.00-2.00 sec  1.21 GBytes  10.4 Gbits/sec  1          5.84 MBytes
[ 7] 2.00-3.01 sec  1.13 GBytes  9.64 Gbits/sec  0          6.07 MBytes
[ 7] 3.01-4.00 sec  1.39 GBytes  12.0 Gbits/sec  1          6.28 MBytes
[ 7] 4.00-5.00 sec  1.45 GBytes  12.5 Gbits/sec  0          6.48 MBytes
[ 7] 5.00-6.00 sec  1.41 GBytes  12.1 Gbits/sec  0          6.62 MBytes
[ 7] 6.00-7.00 sec  1.65 GBytes  14.2 Gbits/sec  0          6.80 MBytes
[ 7] 7.00-8.00 sec  1.81 GBytes  15.6 Gbits/sec  0          6.95 MBytes
[ 7] 8.00-9.00 sec  1.74 GBytes  14.9 Gbits/sec  0          7.08 MBytes
[ 7] 9.00-10.00 sec 1.23 GBytes  10.6 Gbits/sec  0          7.22 MBytes
[ 7] 10.00-10.01 sec 11.1 MBytes  15.8 Gbits/sec
[ ID] Interval      Transfer      Bitrate
[ 7] 0.00-10.01 sec 14.3 GBytes  12.3 Gbits/sec
Server listening on 5201
```

Выполнение лабораторной работы

```

[ 7] 5,00-6,00 sec 1,38 GBytes 11,9 Gbits/sec 0 6,62 MBytes
[ 7] 6,00-7,00 sec 1,65 GBytes 14,1 Gbits/sec 0 6,80 MBytes
[ 7] 7,00-8,00 sec 1,81 GBytes 15,6 Gbits/sec 0 6,95 MBytes
[ 7] 8,00-9,00 sec 1,74 GBytes 14,9 Gbits/sec 0 7,08 MBytes
[ 7] 9,00-10,01 sec 1,25 GBytes 10,7 Gbits/sec 0 7,22 MBytes
-----
[ ID] Interval          Transfer          Bitrate          Retr          sender
[ 7] 0,00-10,01 sec 14,3 GBytes 12,3 Gbits/sec 2
[ 7] 0,00-10,01 sec 14,3 GBytes 12,3 Gbits/sec receiver

iperf Done.
root@mininet-vn:/home/mininet# sudo xqdisc add dev hl-eth0 root tbf rate 10gb
it burst 5000000 limit 15000000
root@mininet-vn:/home/mininet# egrep '^CONFIG_HZ_[0-9]+' /boot/config- `uname -r`
grep: /boot/config-: No such file or directory
grep: 5,4,0-42-generic: No such file or directory
root@mininet-vn:/home/mininet# egrep '^CONFIG_HZ_[0-9]+' /boot/config-`uname -r`
CONFIG_HZ_250=y
root@mininet-vn:/home/mininet# iperf3 -c 10.0.0.2
iperf3: error - unable to connect to server: Connection refused
root@mininet-vn:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 50190 connected to 10.0.0.2 port 5201

```

```

[ 7] 2.00-3.01 sec 1.13 GBytes 9.64 Gbits/sec
[ 7] 3.01-4.00 sec 1.39 GBytes 12.0 Gbits/sec
[ 7] 4.00-5.00 sec 1.45 GBytes 12.5 Gbits/sec
[ 7] 5.00-6.00 sec 1.41 GBytes 12.1 Gbits/sec
[ 7] 6.00-7.00 sec 1.65 GBytes 14.2 Gbits/sec
[ 7] 7.00-8.00 sec 1.81 GBytes 15.6 Gbits/sec
[ 7] 8.00-9.00 sec 1.74 GBytes 14.9 Gbits/sec
[ 7] 9.00-10.00 sec 1.23 GBytes 10.6 Gbits/sec
[ 7] 10.00-10.01 sec 11.1 MBytes 15.8 Gbits/sec
-----
[ ID] Interval Transfer Bitrate
[ 7] 0.00-10.01 sec 14.3 GBytes 12.3 Gbits/sec
-----
Server listening on 5201
-----
^Ciperf3: interrupt - the server has terminated
root@mininet-vx/home/mininet# iperf3 -s
warning: this system does not seem to support IPv6 - trying IPv4
-----
Server listening on 5201
-----
Accepted connection from 10.0.0.1, port 60188
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 60190

```

Выполнение лабораторной работы

```
*host: h1*@mininet-vm
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
iperf3: error - unable to connect to server: Connection refused
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 60190 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer      Bitrate      Retr  Cwnd
[ 7] 0.00-1.00 sec  1.09 GBytes  9.37 Gbits/sec  0    8,24 MBytes
[ 7] 1.00-2.00 sec  1.11 GBytes  9.51 Gbits/sec  0    8,24 MBytes
[ 7] 2.00-3.00 sec  1.10 GBytes  9.48 Gbits/sec  0    8,24 MBytes
[ 7] 3.00-4.00 sec  1.09 GBytes  9.40 Gbits/sec  0    8,24 MBytes
[ 7] 4.00-5.00 sec  1.11 GBytes  9.55 Gbits/sec  0    8,24 MBytes
[ 7] 5.00-6.00 sec  1.11 GBytes  9.53 Gbits/sec  0    8,24 MBytes
[ 7] 6.00-7.00 sec  1.11 GBytes  9.53 Gbits/sec  0    8,24 MBytes
[ 7] 7.00-8.00 sec  1.12 GBytes  9.58 Gbits/sec  0    8,24 MBytes
[ 7] 8.00-9.00 sec  1.11 GBytes  9.55 Gbits/sec  0    8,24 MBytes
[ 7] 9.00-10.00 sec 1.11 GBytes  9.57 Gbits/sec  0    8,24 MBytes
-----
[ ID] Interval      Transfer      Bitrate      Retr
[ 7] 0.00-10.00 sec 11.1 GBytes  9.51 Gbits/sec  0
[ 7] 0.00-10.00 sec 11.0 GBytes  9.48 Gbits/sec
sender
receiver

iperf Done.
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
root@mininet-vm:/home/mininet#
```

```
*host: h2*@mininet-vm
Server listening on 5201
Accepted connection from 10.0.0.2 port 5201 connected to 10.0.0.1 port 60190
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 60190
[ ID] Interval      Transfer      Bitrate
[ 7] 0.00-1.00 sec  1.08 GBytes  9.26 Gbits/sec
[ 7] 1.00-2.00 sec  1.10 GBytes  9.49 Gbits/sec
[ 7] 2.00-3.00 sec  1.10 GBytes  9.49 Gbits/sec
[ 7] 3.00-4.00 sec  1.10 GBytes  9.41 Gbits/sec
[ 7] 4.00-5.00 sec  1.11 GBytes  9.57 Gbits/sec
[ 7] 5.00-6.00 sec  1.09 GBytes  9.37 Gbits/sec
[ 7] 6.00-7.01 sec  1.12 GBytes  9.60 Gbits/sec
[ 7] 7.01-8.00 sec  1.12 GBytes  9.65 Gbits/sec
[ 7] 8.00-9.00 sec  1.11 GBytes  9.50 Gbits/sec
[ 7] 9.00-10.00 sec  1.10 GBytes  9.49 Gbits/sec
[ 7] 10.00-10.00 sec  896 KBytes  2.19 Gbits/sec
-----
[ ID] Interval      Transfer      Bitrate
[ 7] 0.00-10.00 sec 11.0 GBytes  9.48 Gbits/sec
receiver

Server listening on 5201
```


Выполнение лабораторной работы

```
switch: s1" (root@mininet-vm)
MAX_BYTES]]
root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 root handle 1: netem delay 10ms
root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 parent 1: handle 2: tbf rate 2gbit burst 1000000 limit 20000
00
root@mininet-vm:/home/mininet#
```

```
host: h1" @mininet-vm
[ 7] 3,00-4,00 sec 1.11 GBytes 9.55 Gbits/sec 0 8,02 MBytes
[ 7] 4,00-5,00 sec 1.11 GBytes 9.52 Gbits/sec 0 8,02 MBytes
[ 7] 5,00-6,00 sec 1.11 GBytes 9.57 Gbits/sec 0 8,02 MBytes
[ 7] 6,00-7,00 sec 1.12 GBytes 9.60 Gbits/sec 0 8,02 MBytes
[ 7] 7,00-8,00 sec 1.11 GBytes 9.57 Gbits/sec 0 8,02 MBytes
[ 7] 8,00-9,00 sec 1.11 GBytes 9.55 Gbits/sec 0 8,02 MBytes
[ 7] 9,00-10,00 sec 1.11 GBytes 9.57 Gbits/sec 0 8,02 MBytes
-----
[ ID] Interval      Transfer      Bitrate      Retr
[ 7] 0,00-10,00 sec 11.1 GBytes 9.55 Gbits/sec 0
[ 7] 0,00-10,02 sec 11.1 GBytes 9.53 Gbits/sec 0
sender
receiver

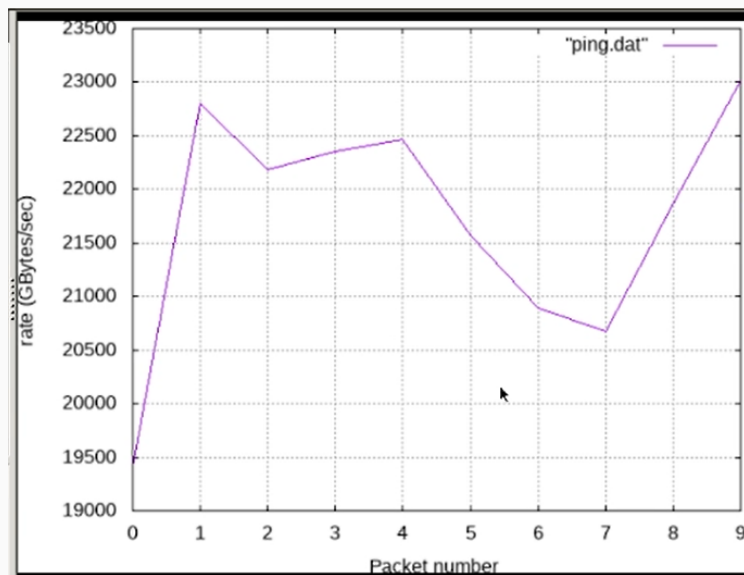
iperf Done.
root@mininet-vm:/home/mininet# ping -c 4 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=12.3 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=11.3 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=10.2 ms
--- 10.0.0.2 ping statistics ---
664 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 10.209/11.126/12.327/0.797 ms
root@mininet-vm:/home/mininet#
```

```
host: h2" @mininet-vm
Server listening on 5201
Accepted connection from 10.0.0.1, port 60192
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 60194
[ ID] Interval      Transfer      Bitrate
[ 7] 0,00-1,00 sec 1.10 GBytes 9.45 Gbits/sec
[ 7] 1,00-2,01 sec 1.11 GBytes 9.46 Gbits/sec
[ 7] 2,01-3,00 sec 1.10 GBytes 9.50 Gbits/sec
[ 7] 3,00-4,00 sec 1.11 GBytes 9.56 Gbits/sec
[ 7] 4,00-5,00 sec 1.12 GBytes 9.59 Gbits/sec
[ 7] 5,00-6,01 sec 1.11 GBytes 9.48 Gbits/sec
[ 7] 6,01-7,00 sec 1.12 GBytes 9.63 Gbits/sec
[ 7] 7,00-8,00 sec 1.11 GBytes 9.54 Gbits/sec
[ 7] 8,00-9,00 sec 1.11 GBytes 9.54 Gbits/sec
[ 7] 9,00-10,00 sec 1.12 GBytes 9.61 Gbits/sec
[ 7] 10,00-10,02 sec 8.95 MBytes 5.43 Gbits/sec
-----
[ ID] Interval      Transfer      Bitrate
[ 7] 0,00-10,02 sec 11.1 GBytes 9.53 Gbits/sec
sender
receiver

Server listening on 5201
iperf3: interrupt - the server has terminated
```

```
*** Starting CLI:
mininet> exit
*** Stopping network
*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
sudo chown mininet:mininet ping.dat
gnuplot ping_plott
sudo chown mininet:mininet ping.png
mininet@mininet-vm:~/work/lab_netem_ii/simple-drop$ ls
lab_netem_iii.py  lab_netem_ii.py.save  nano.save  ping_plott
lab_netem_ii.py   Makefile              ping.dat   ping.png
mininet@mininet-vm:~/work/lab_netem_ii/simple-drop$ geeqie ping.da
mininet@mininet-vm:~/work/lab_netem_ii/simple-drop$ geeqie ping.pn
█
```

Рис. 7: Скрипт для воспроизводимого эксперимента



В результате выполнения данной лабораторной работы я познакомилась с принципами работы дисциплины очереди Token Bucket Filter, которая формирует входящий/исходящий трафик для ограничения пропускной способности, а также получила навыки моделирования и исследования поведения трафика посредством проведения интерактивного и воспроизводимого экспериментов в Mininet.